Central Bank Balance Sheet Growth and Subsequent US Equity Returns

SIQuant Research[1]

[1]Hypothesis × Quant

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Abstract

Hypothesis: H_0: the four-week change in Federal Reserve total assets has no predictive power for the S&P 500 return over the next 20 sessions; H_1: the slope is positive

Data: 519 weekly observations, 2016-10-24 to 2026-09-25; information date set to the Friday after the Thursday release

Method: OLS with Newey–West HAC errors (8 lags) and a 2,000-draw circular-shift placebo; the decision needs both one-sided $p < 0.05$

Results: $\beta = 0.256$ (HAC $t = 4.53$, $p = 0.0000$; placebo $p = 0.0085$), $R^2 = 0.0288$; mean forward return 0.94% after expansion and 1.05% after contraction; without 2020 the slope is 0.262 ($t = 1.19$).

Research decision: ACCEPTED

1 Design

Liquidity provision by central banks is often linked to risk-asset performance [Lewis and Roth, 2019]. The H.4.1 release is public on Thursday afternoon, so the earliest tradable close is Friday; the outcome window starts from that close. The signal is the four-week log change in WALCL.

2 Results

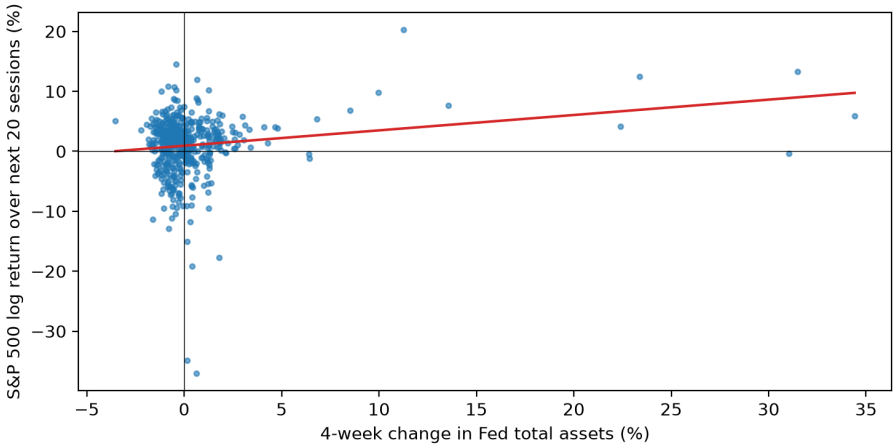


Figure 1: Four-week balance-sheet change against the next 20-session S&P 500 return.

Excluding the 2020 episode the slope is 0.262 (HAC $t = 1.19$); mean 20-session returns after expansion and contraction weeks are 0.94% and 1.05%, so the sign of the change alone carries little information and the pooled slope depends on large balance-sheet moves.

3 Limitations

The sample contains one expansion (2020–21) and one contraction (2022–) regime, so effective independent observations are very few. Balance-sheet growth is endogenous to market stress. Reserve and Treasury General Account dynamics are not modelled.

References

Vivien Lewis and Markus Roth. Monetary policy and asset prices: the role of central bank balance sheets. *Journal of International Money and Finance*, 96:238–251, 2019.